

IMO Training 2018–19

Phase 2 (Level 1)

Algebra

2018/11/10

A. Quadratic Functions

1. How do you minimise or maximise a quadratic function of one variable?

2. How about quadratic functions in two variables? Illustrate with

$$f(x, y) = x^2 + 2xy + 3y^2 - 4x - 12y.$$

Use as many different ways as possible to minimise the above function.

B. Factor Theorem

3.
 - (a) State the factor theorem.
 - (b) State a stronger form of the factor theorem in case a polynomial has integer coefficients.

4. Let $f(x)$ be a polynomial with integer coefficients. Show that each of the following is a sufficient condition for the equation $f(x) = 0$ to have no integer solution.
 - (a) There exist distinct integers a, b, c such that $|f(a)| = |f(b)| = |f(c)| = 1$.
 - (b) There exist distinct integers a, b, c, d, e such that $f(a) = f(b) = f(c) = f(d) = f(e) = 7$.
 - (c) Both $f(0)$ and $f(1)$ are odd.

C. Divisibility of Polynomials

5. Show that $x^{10} + x^5 + 1$ can be factorised into the product of two non-constant polynomials with integer coefficients.
6. Find all natural numbers n for which $1 + x + x^2 + \cdots + x^n$ divides $1 + x^2 + x^4 + \cdots + x^{2n}$.
7. Find all polynomials $f(x)$ for which $x^2 + 1$ divides $f(x)$ and $x^3 + x^2 + 1$ divides $f(x) + 1$.

D. Fundamental Theorem of Algebra

8. What does the Fundamental Theorem of Algebra say?
9. Find all polynomials $p(x)$ for which $p(0) = 0$ and $p(x^2 + 1) = [p(x)]^2 + 1$ for all x .
10. Find all polynomials $f(x)$ with real coefficients such that
- $$f(x)f(x+1) = f[x(x+1)+1]$$
- for all real numbers x .